

Algorithms and Datastructures — Exercise Sheet 5

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Exercise 1: Bad Hash Functions

1. Keys 0-9 have 0; Keys 10-19 have 1... Which is **not uniform**
2. All the even Buckets are **wasted**
3. Out of the hash values: When $m=10, x=0$ or 9 . $x \bmod m + \frac{m}{x+1} = 10$ **Out of Range**
4. This hash function is **not deterministic**.
5. x can not be 0
6. Single hash function is already good. It wastes Resources.

Exercise 2: (No) Families of Universal Hash Functions

1. $h(0) = h(m) = 0$ Hence the hash function is not c-universal.
2. Assume $\sum_0^k a_i x_i = \sum_0^k a_i y_i \bmod m$
means $a_j x_j - a_j y_j = \text{Constant} \bmod m$
 $a_j (x_j - y_j) = \text{Constant} \bmod m$
 $x_j - y_j = \frac{\text{Constant}}{a_j} \bmod m$
 $\frac{\text{Constant}}{a_j} = 1$
So only one a_j and $a \in \{1, \dots, m-1\}$, so 1-universal